

Week 5 Tutorial

Introduction

This tutorial will help you familiarize yourself with optimal portfolio selection. We will use the **Solver** add-in from Microsoft Excel, so please:

- open the [W5Q1Solver.xlsx](#) file from Moodle.
- install Solver

Question 1

Assume three risky assets, namely Microsoft (M), Nordstrom (N) and Starbucks (S). Let r_i denote the return on asset i where $i = M, N, S$. The following information is given for these assets:

Stock i	μ_i	σ_i	Pairs (i, j)	σ_{ij}
M (Microsoft)	0.0427	0.1000	(M,N)	0.0018
N (Nordstrom)	0.0015	0.1044	(M,S)	0.0011
S (Starbucks)	0.0285	0.1411	(N,S)	0.0026

where μ_i , σ_i and σ_{ij} denote expected returns, standard deviations and covariances for each asset, respectively.

1. Construct a 3×1 vector, μ , containing the expected returns of these three assets. Similarly, construct the 3×3 variance-covariance matrix, Σ , associated with the performance of these assets based on the information provided (do this in the **W5Q1Solver.xlsx** file).
2. Consider a portfolio, P, that consists of assets M,N,S, with return given by:

$$r_P = w_M r_M + w_N r_N + w_S r_S,$$

where w_M, w_N, w_S are the (non-negative) weights assigned to each of the assets, such that $w_M + w_N + w_S = 1$. Using matrix notation write the analytical expressions for portfolio expected return, μ_P , and portfolio variance, σ_P^2 .

3. Find the numerical solution to this optimization problem in Excel using the information provided in the table above (make use of the Excel add-in “Solver”).

Fill in the blanks of the Q1 - solver tab in the **W5Q1Solver.xlsx** file from Moodle. You can also use the Solver and Matrix algebra handout as a resource.

4. Find the analytic solution to the following optimization problem

$$\min_{w_M, w_N, w_S} \sigma_P^2, \text{ such that } w_M + w_N + w_S = 1,$$

using the Lagrange multiplier method. [not required]

5. Using the analytical formula for the vector of optimal weights in part 4, compute the optimal weights assigned to the Global Minimum Variance portfolio.

Use the tab named 'Analytical' in the Excel file (as in Part 3) and matrix multiplication in Excel.

Question 2

Consider stock prices for Microsoft (MSFT) and Walmart (WMT) from 1990 to 2024. The stock prices can be found in the file [msft__wmt.wfl](#) in the tutorial folder. Use these data to answer the following questions.

1. Compute the log-returns on Microsoft and Walmart at the monthly frequency.

Download the "msft__wmt.xlsx" file from Moodle into your project and run the code below.

First we need to load the `xts` library and create better dates (the original date format is incompatible with `xts`). We will look at what we have and plot the data.

```
library(xts)
```

```
Loading required package: zoo
```

```
Attaching package: 'zoo'
```

```
The following objects are masked from 'package:base':
```

```
as.Date, as.Date.numeric
```

```

mw_df <- readxl::read_excel("msft_wmt.xlsx") # <-- change filename

# ignore the date column
mw_df = mw_df[, -1]

# Jan 1990 to March 2024
# create an xts object, but first create the date.
idx <- seq(from = as.Date("1990-01-01"), by = "month", length.out = nrow(mw_df))
prices.xts <- xts(mw_df, order.by = idx)

# just display month-year without date
index(prices.xts) <- as.yearmon(index(prices.xts))
head(prices.xts)

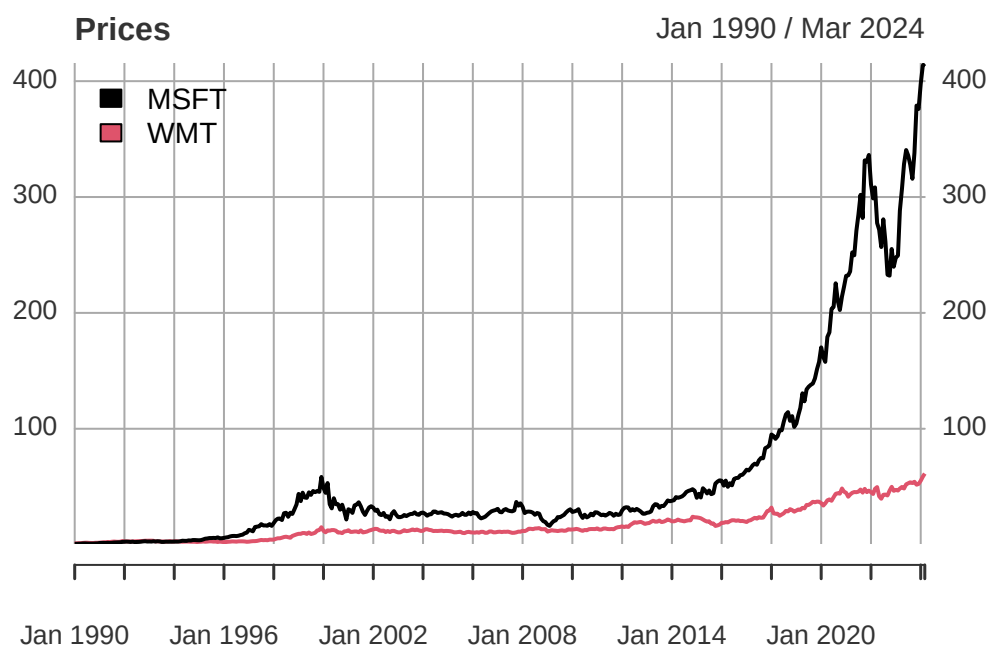
```

	MSFT	WMT
Jan 1990	0.642361	1.093507
Feb 1990	0.685764	1.128781
Mar 1990	0.769097	1.212157
Apr 1990	0.805556	1.274974
May 1990	1.013889	1.448397
Jun 1990	1.055556	1.602550

```

plot(prices.xts, legend.loc = "topleft", main = "Prices")

```



Now, as usual, we calculate log-returns. This time we will do both at a time and create two columns.

```
# log return for both msft and wmt
logret.xts <- diff(log(prices.xts))
logret.xts <- na.omit(logret.xts) # removes the first NA created by diff()
colnames(logret.xts) <- c("msft", "wmt") # name the column
```

2. Compute the sample variance-covariance matrix of returns for these two stocks.

Since we have 2 columns of data, we have a simple formula to get the sample var-cov matrix.

3. Estimate the regression model

$$r_{wal,t} = \beta_0 + \beta_1 (r_{wal,t} - r_{mic,t}) + \epsilon_t,$$

where ϵ_t is a disturbance term. Interpret the estimate of β_1 .

We can use logret.xts directly. We will create a new variable for the difference - msopt to make the output cleaner. We could use the I(.) function to create the difference directly in the lm(.) function, but the output is messy.

```
msopt <- logret.xts$wmt-logret.xts$msft
minvar <- lm(logret.xts$wmt ~ msopt)
summary(minvar)
```

Call:

```
lm(formula = logret.xts$wmt ~ msopt)
```

Residuals:

Min	1Q	Median	3Q	Max
-0.227714	-0.034376	-0.000255	0.032647	0.189722

Coefficients:

	Estimate	Std. Error	t value	Pr(> t)
(Intercept)	0.011546	0.002769	4.169	3.74e-05 ***
msopt	0.289473	0.030860	9.380	< 2e-16 ***

Signif. codes: 0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.05595 on 408 degrees of freedom

Multiple R-squared: 0.1774, Adjusted R-squared: 0.1754

F-statistic: 87.99 on 1 and 408 DF, p-value: < 2.2e-16

5. Using the regression results from the previous question, construct a hypothesis test that an equally weighted portfolio would result in a minimum variance portfolio

$$H_0 : \beta_1 = 0.5 \text{ v.s. } H_0 : \beta_1 \neq 0.5.$$

NOTE: the calculated t-value and the p-value from the regression output is based upon testing if $\beta_1 = 0$ against $\beta_1 \neq 0$. This means that we have to calculate the t-value manually, as well as the p-value.

We test $H_0 : \mu = 0.5$ against $H_1 : \mu \neq 0.5$. The test statistic is $t = (\hat{\beta}_1 - 0.5)/s.e.(\hat{\beta}_1) \sim t_{408}$ under H_0 . The calculated t-statistic is $t = (0.289473 - 0.5)/0.030860$. The code below does this in R - you must fill in some gaps and set `eval = TRUE`.

```
# Find the calculate t-value
```

```
bhat <- coef(minvar)["msopt"]
```

```
se <- summary(minvar)$coefficients["msopt", "Std. Error"]
```

```
t_calc <- ???
```

```
c("t-calc", round(t_calc, 3))
```

```
# Find the p-value
```

```
# degrees of freedom = n-2
```

```
dfr <- nobs(minvar) - 2
```

```
# p-value - find area in tails from t_calc in a t dist with n-1 df
```

```
# pt gives area to the LEFT, so we need to always use the negative t_calc
```

```
# -abs(.) will always be negative, regardless of the sign of t_calc
```

```
p_val <- round(???, dfr),3)
```

```
c("p-value", p_val)
```